

CHATGPT UNLEASHED



LOGISTICS



Class Hours:

- **Class will be 10-5 EST**
- **Lunch at 1pm for 75 minutes**
- **20-minute break at 11:30 and 3:00**



Telecommunication:

- **Turn off or set electronic devices to silent (not vibrate)**
- **Reading or attending to devices can be distracting to other students**
- **Try to delay until breaks or after class**

INSTRUCTOR : DANIEL BLANCO

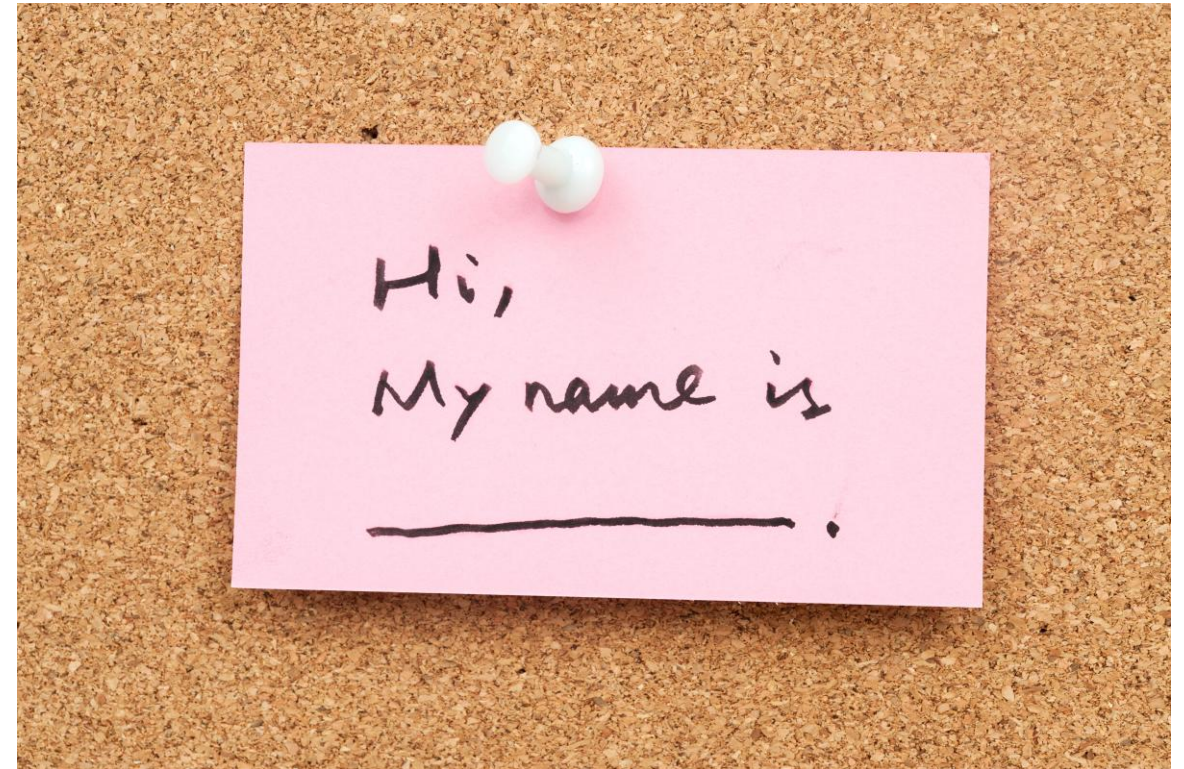
About

- 15+ years of experience, CIS, MBA
- Started in database development (telecom systems)
- Transitioned from development to operations in a health-tech startup
- Experience supporting front-office and administrative functions in financial services
- Led bootcamps at Rutgers and Pepperdine
- 5+ years of corporate IT training experience
- Based in Utah | Father of 4

INTRODUCE YOURSELF

Time to introduce yourself:

- Name
- What is your role in the organization
- Indicate AI experience
- Expectations for this training



WHAT THIS COURSE COVERS

This is a three-day, hands-on course. You will move from asking ChatGPT simple questions to using it as a practical workspace for research, analysis, content creation, and software development.

- Day 1 covers how LLMs work, prompt engineering, multimodal inputs, deep research, and ChatGPT Projects.
- Day 2 covers tools, connected applications, Skills, and advanced multi-step workflows.
- Day 3 covers Codex, AI governance, and a capstone project.

THREE DAYS AT A GLANCE

- Day 1:** Mastering ChatGPT and Modern AI
- Day 2:** ChatGPT Tools, Projects, Skills, and Workflows
- Day 3:** Codex, AI Safety, and the Capstone



THE GOAL

By the end of this course, you will be able to apply ChatGPT to real business and technical problems. The focus is on building practical capability, not memorizing theory.

You will leave with concrete workflows, reusable prompts, and a clear sense of where AI adds value and where human judgment remains essential.

HOW THE LABS WORK

Each lab follows the same pattern. You are given a business scenario, a set of objectives, and a starting point. You do the work.

- Labs are hands-on, not demonstrations
- Some labs are fully guided with example prompts to run
- Later labs give you the goal and let you decide the approach
- Lab difficulty increases across the three days

HOW TO GET THE MOST FROM THIS COURSE

Come in with a real task from your own work. Every lab has space to apply what you are learning to your actual situation, not just the provided scenario.

The skills compound. A well-structured prompt from Day 1 becomes the foundation for the Skills and Codex workflows in Days 2 and 3. Each lab builds on the last.



WHAT IS GENERATIVE AI?

Generative AI refers to models that create new content based on patterns learned from vast amounts of existing data. ChatGPT is one of the most capable of these systems available today.

Unlike traditional software that follows explicit rules, generative AI learns patterns from data and uses them to generate responses. This makes it flexible, but also unpredictable in ways traditional software is not.

HOW LLMS LEARN

A large language model is trained by processing enormous quantities of text. During training it learns which words and phrases tend to follow which other words and phrases.

After training, the model does not look anything up. It generates responses by predicting the most statistically likely continuation of whatever you have given it. This is why the same question can produce different answers on different days.





Training Data

A large collection of text, books, articles, web pages, and more.



Large Language Model

Learns patterns, relationships, and knowledge from the data.



Your Prompt

You provide a question, request, or instruction in natural language.

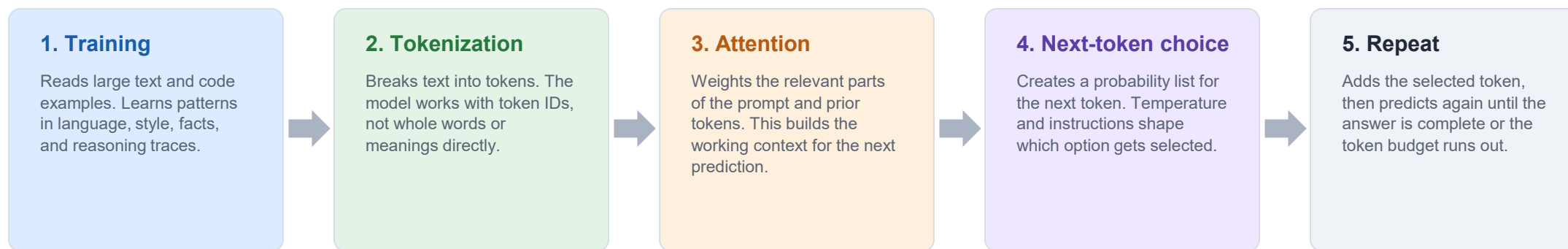


Generated Response

The model generates a helpful, relevant response based on what it learned.

How an LLM turns a prompt into an answer

The model repeatedly predicts the next token. The useful trick is the context it uses for each prediction.



What this means in practice

Good prompts improve the context used for prediction. They tell the model what role to play, which facts matter, what output shape to use, and what uncertainty to flag.

Main failure mode: a fluent continuation can still be wrong. Ask for sources, constraints, checks, and assumptions when accuracy matters.

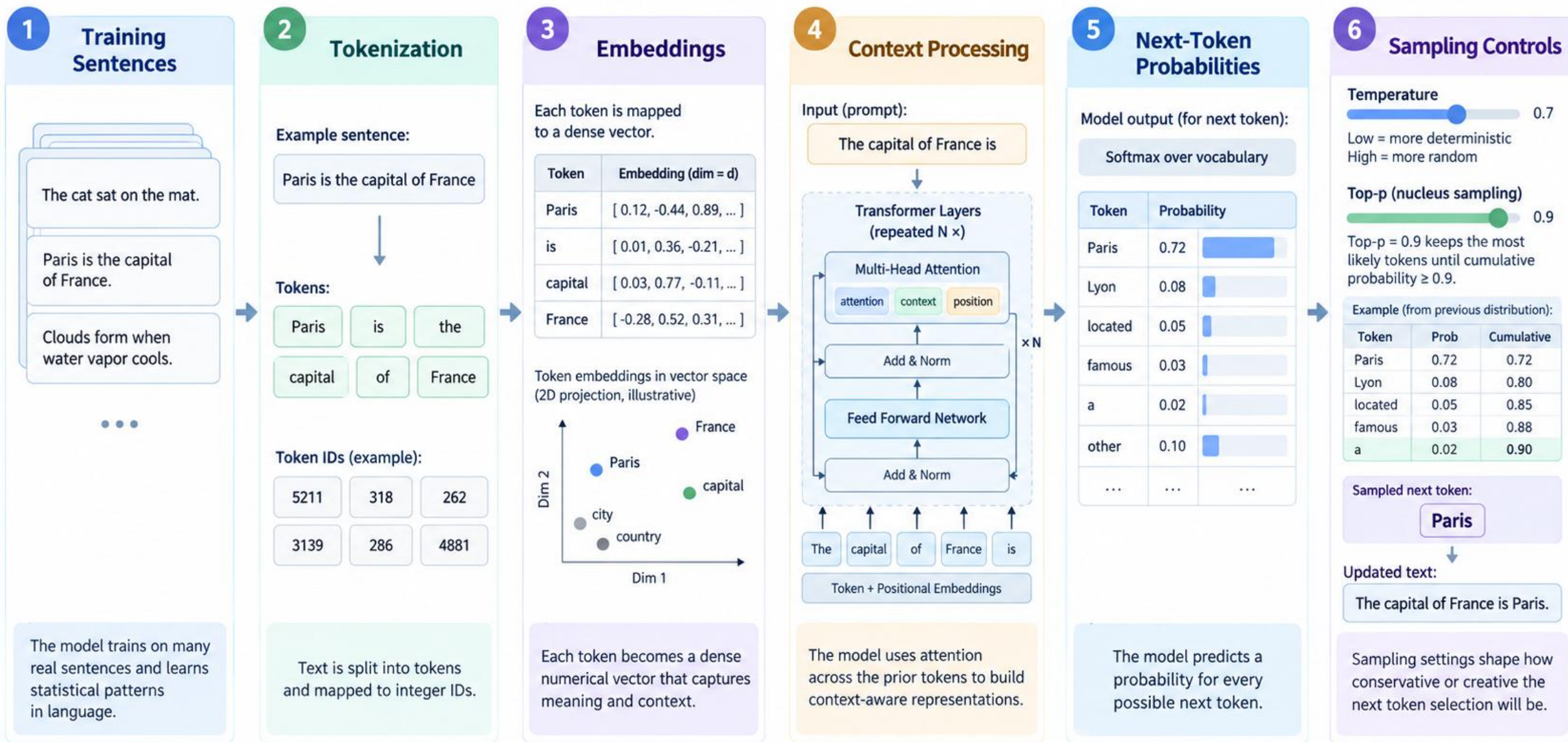
WHAT A TOKEN IS

LLMs do not process text word by word. They break text into units called tokens, which are roughly equivalent to word fragments or short words. "Unbelievable" might be three tokens. "AI" might be one.

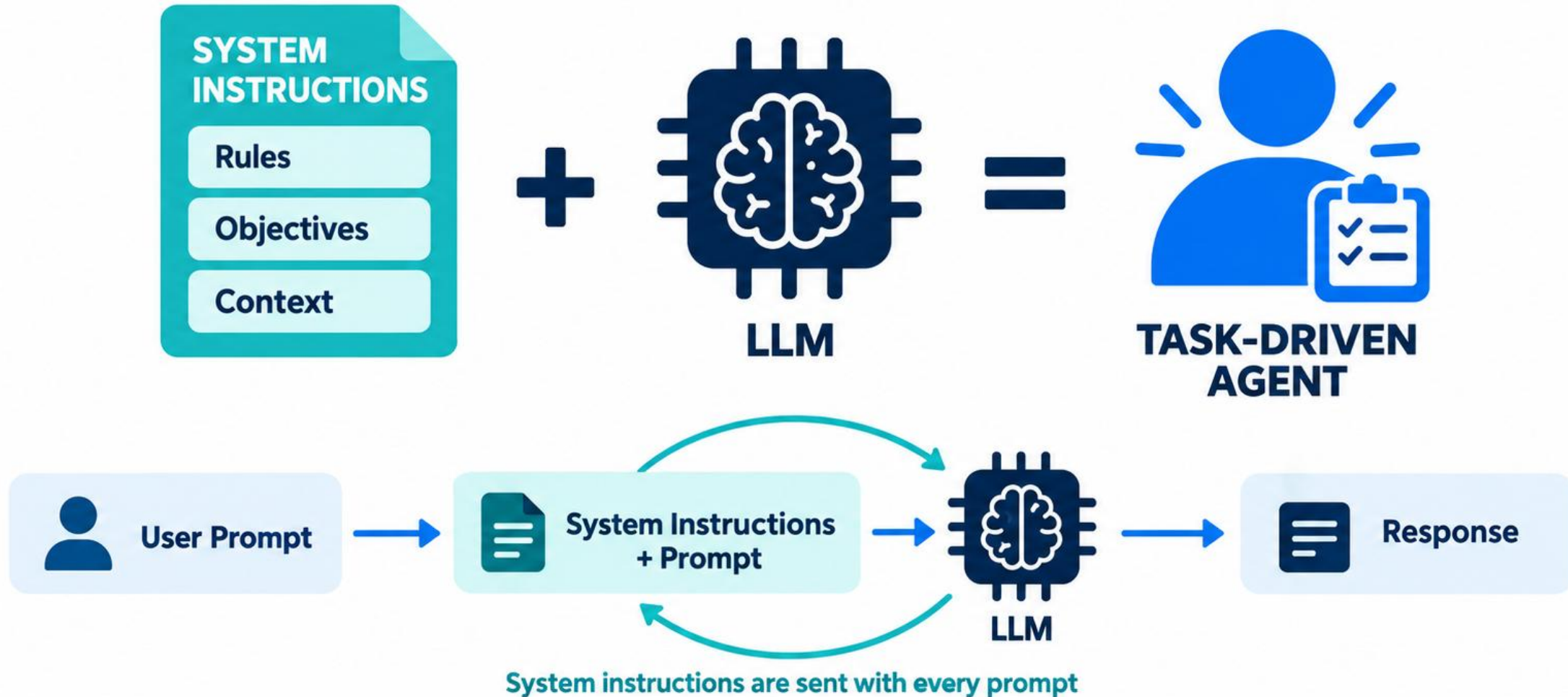
Tokens matter because everything in a ChatGPT conversation is measured in them. Your prompt, the response, uploaded files, and system instructions all **consume tokens from the same shared budget**.

How a Language Model Generates the Next Token

From real text training data to tokenization, embeddings, probability distributions, and sampling



How System Instructions Shape an Agent



LLMs are not deterministic. Instructions guide behavior, but cannot guarantee the rules will always be followed.

CONTEXT WINDOWS

Every ChatGPT conversation has a limit on how many tokens it can hold at once. This is called the context window.

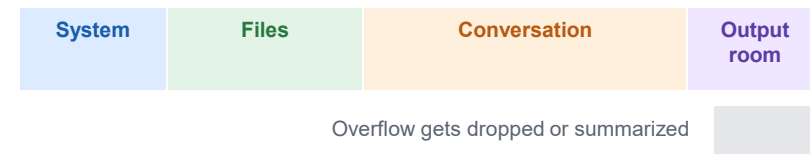
When a conversation exceeds the context window, older content is dropped. Very long conversations, large file uploads, and lengthy instructions all push you toward that limit faster than short ones.

WHAT FILLS THE CONTEXT WINDOW

- System and project instructions (loaded at the start of every conversation)
- Uploaded files (converted to text and included)
- Your prompts and all of ChatGPT's responses so far
- Web search results retrieved during the conversation
- Any images or data passed as input

Context window

Everything the model can use right now



What competes for space

System rules, project instructions, chat history, retrieved file chunks, tool results, and the answer itself.

Why details disappear

Large files are retrieved selectively. Old messages can fall out of view when the conversation grows.

How to control it

Restate critical facts, keep files focused, and put durable behavior in project instructions.

CONTEXT WINDOWS

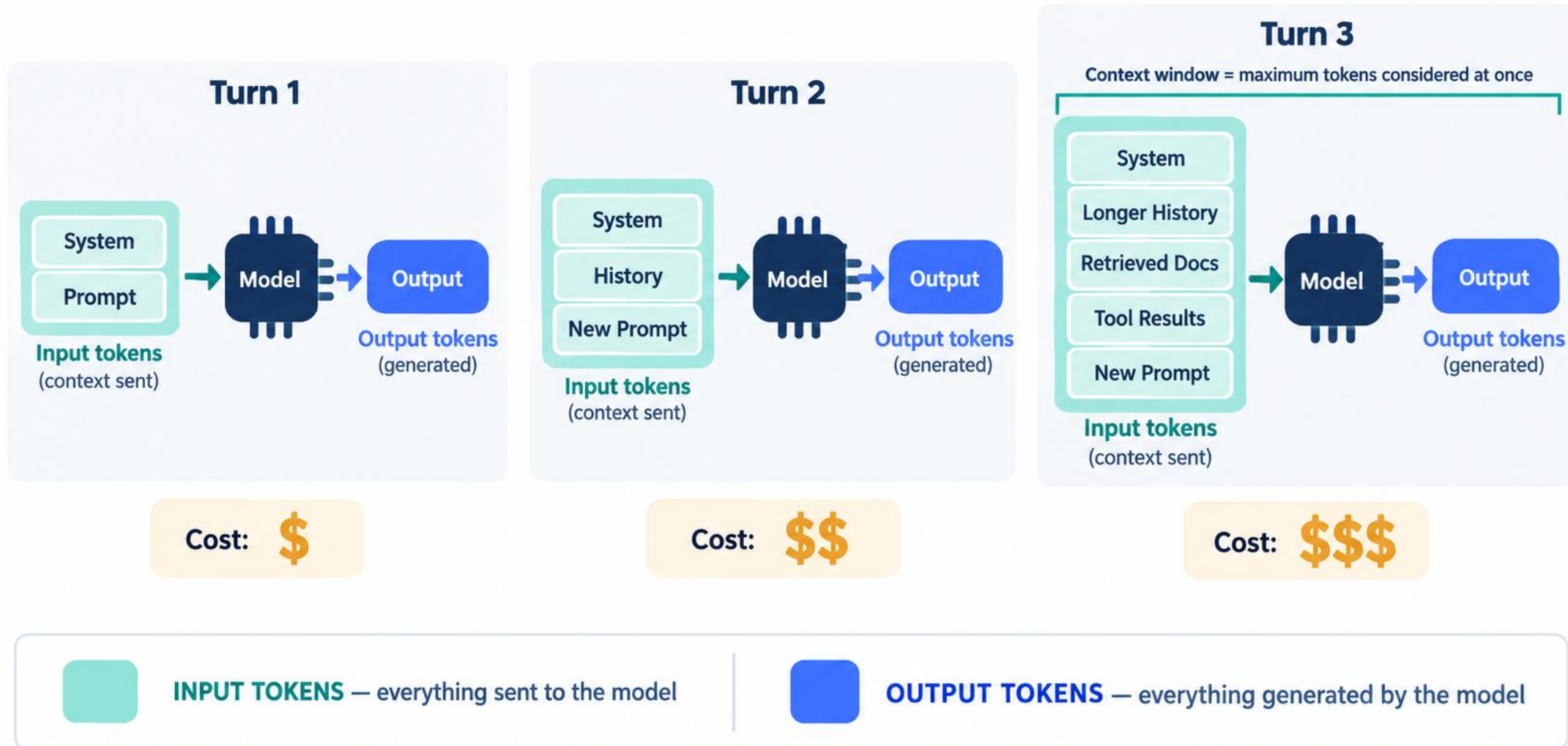
Token usage is tied directly to cost. Providers usually price tokens separately as **input tokens** and **output tokens**:

Input tokens are everything sent to the model: your latest prompt, system instructions, chat history, retrieved documents, tool results, and any other context.

Output tokens are what the model generates in response.

In a chat-style application, each new prompt typically sends the conversation history again as input. So as the context grows, every follow-up can become more expensive because the model is re-reading more prior context each time. The context window is the maximum amount the model can consider at once, but using more of it also increases latency and cost.

More Context = More Tokens = More Cost



MANAGING CONTEXT WINDOWS

You can see the percentage of the context window being used inside ChatGPT. Each model has a maximum context window: the total amount of conversation, instructions, files, and generated output it can consider at one time.

When the conversation grows beyond that limit, older parts of the chat may be dropped, summarized, or no longer available to the model.

Pro Tip: Starting a new chat can reduce unnecessary token usage because the model no longer has to process the full history of a long conversation. You can also use `/compact` to summarize the current conversation into a shorter form, preserving the important details while freeing up space in the context window.

NONDETERMINISTIC BEHAVIOR

ChatGPT is not deterministic. Send the same prompt twice and you may get two different answers. This is by design.

A setting called temperature controls how much randomness the model introduces when generating tokens. Higher temperature produces more creative and varied output. Lower temperature produces more consistent and conservative output. Most ChatGPT tasks run with a moderate temperature.

THE SAME PROMPT, DIFFERENT ANSWERS

This behavior is a feature, not a bug. It produces variety in creative tasks and avoids getting stuck in repetitive patterns.

It is a problem when you need exact reproducibility, such as in compliance workflows, auditable outputs, or code generation where consistency matters. In those cases, structure your prompt more tightly and review every output.

TWO TYPES OF MODELS

- **Conversational models** respond quickly, optimized for dialogue and content generation
- **Reasoning models** spend time thinking step by step before responding, designed for complex multi-part problems
- Most everyday ChatGPT tasks use conversational models by default
- Reasoning models are slower and cost more tokens but make fewer logical errors on hard tasks
- You can select the model in the ChatGPT interface before starting a conversation



REASONING MODELS IN PRACTICE

Use a reasoning model when the task requires multiple sequential steps of logic, when a wrong intermediate step would invalidate the rest of the answer, or when the problem has strict constraints that must all be satisfied at once.

Use a conversational model when you need a fast response, when the task is content generation or summarization, or when you plan to iterate quickly across many turns.

Newer GPT models are built to support reasoning, but they do not always reason at full intensity. The model can use different reasoning effort levels, such as low, medium, or high. Lower effort is faster and cheaper for simple conversational tasks. Higher effort is better for complex work like coding, analysis, math, debugging, and planning.

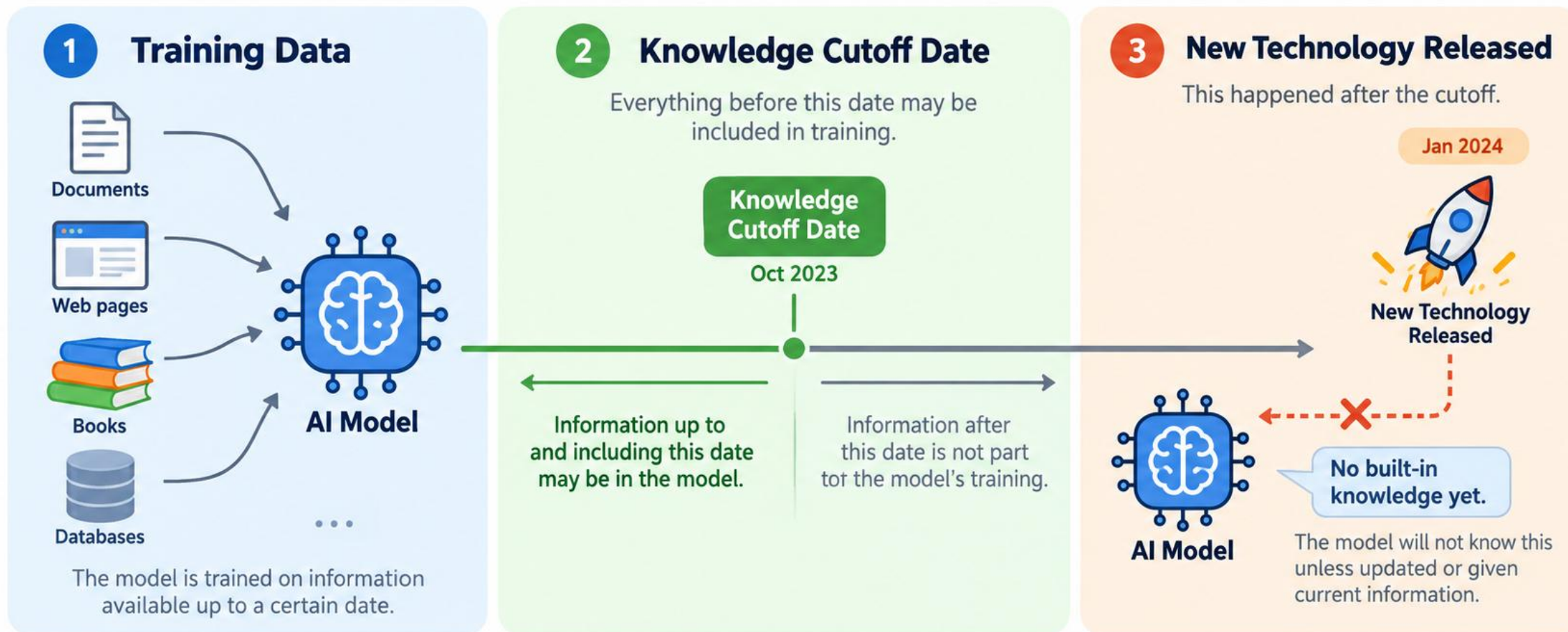
CHATGPT'S TRAINING CUTOFF

ChatGPT was trained on data up to a specific date. It has no knowledge of events, publications, or changes that occurred after that cutoff unless you use web search.

This matters more than most people expect. A regulation passed last quarter, a competitor announcement last month, or a product change last week are all invisible to ChatGPT without current information being provided.

How a Knowledge Cutoff Works

An AI model can only know information it was trained on.



To know about events after the cutoff date, the model needs newer training or access to current information.

This can happen through a new model version or tools like web browsing, APIs, or other external sources.

WHAT CHATGPT DOES NOT HAVE ACCESS TO

- Your organization's internal documents and data (unless you upload them)
- Real-time information of any kind (unless web search is enabled)
- Events and developments after the training cutoff
- Anything you have discussed in previous, unrelated conversations
- Other users' conversations or uploaded files
- Training on private data:** Business accounts Team, Enterprise, and API users are opted out by default, meaning their data is never used for training.

HALLUCINATIONS

A hallucination is a confident, plausible-sounding claim that is factually wrong. This happens because the model predicts likely text rather than looking facts up.

The output reads like knowledge. It can include specific dates, names, page numbers, and citations. All of those details may be invented. Hallucinations are most dangerous when they match what you already expect to be true.



WHY HALLUCINATIONS HAPPEN

The model's job is to produce the most statistically likely continuation of your prompt. If a confident, detailed answer is more likely than an admission of uncertainty, the model may produce that confident answer even when it does not have reliable information.

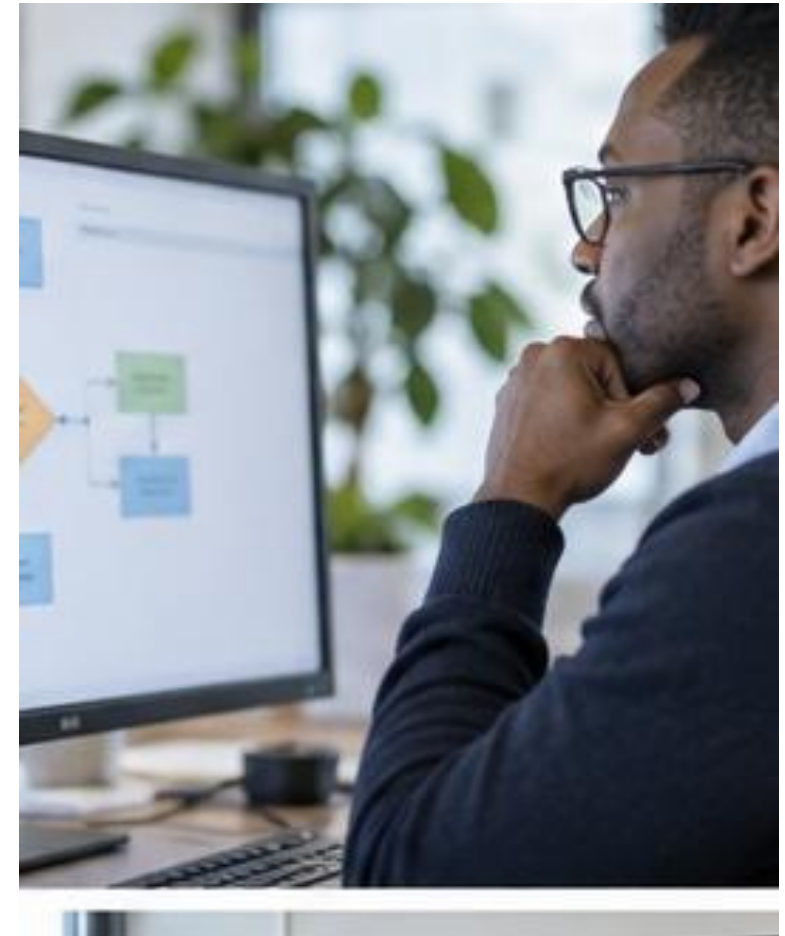
The model is not lying. It has no intent. It is completing patterns, and sometimes those patterns lead it to plausible-sounding conclusions that are not grounded in fact.

SPOTTING AND HANDLING HALLUCINATIONS

- Be most skeptical of specific statistics, citations, names of people, and niche facts
- Any claim that seems surprisingly convenient or that you cannot verify should be checked
- Ask ChatGPT to state its confidence or note where it may be uncertain
- Verify any claim before using it in a deliverable presented to others
- Web search and document upload reduce hallucination risk by grounding responses in real sources

CHOOSING THE RIGHT TOOL

- Content generation, summarization, or drafting: conversational model, no special tools
- Questions that require current information: add web search
- Complex logic or multi-step reasoning: switch to a reasoning model
- Analysis of your own data: upload the file and use the code interpreter
- Open-ended research with many sources: use deep research mode
- Tasks that recur weekly or monthly: build a Skill or Project (covered in Days 2 and 3)



WHY PROMPTS MATTER

The same underlying model produces very different outputs depending on what you give it. A poorly constructed prompt produces generic output because ChatGPT has to guess what you actually want.

Prompt engineering is not about tricks or magic phrases. It is about providing enough information and structure that the model can do the task the way you would want a smart colleague to do it.

THE FOUR ELEMENTS OF A STRUCTURED PROMPT

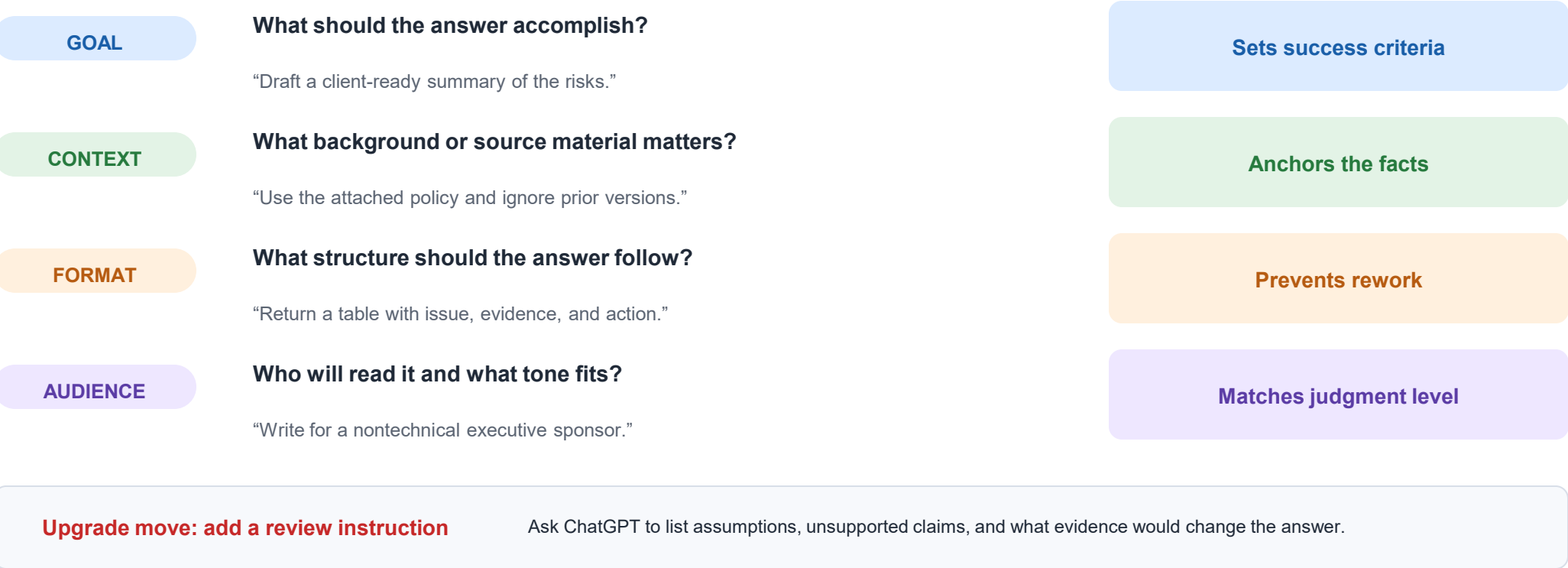
- Goal:** What you want the output to accomplish
- Context:** The situation, background, and relevant facts
- Format:** The shape of the response (length, structure, tone)
- Audience:** Who will read or use the result

Most weak prompts are missing at least two of these four. Adding them is the single highest-leverage improvement you can make.



Anatomy of a structured prompt

A strong prompt reduces guesswork by making the task, source material, output shape, and review standard explicit.



A PROMPT BEFORE AND AFTER

A vague prompt says: "Write a summary of our Q3 project." ChatGPT guesses the project, the audience, the length, and the format.

A structured prompt says: "You are helping a project manager at a financial services firm. Goal: Write a 150-word executive summary of our Q3 technology modernization project. Context: The project ran July through September, came in 5% under budget, and met all three milestones. One feature was deferred to Q4. Audience: Senior leadership with non-technical backgrounds. Format: Three paragraphs covering accomplishments, budget, and next steps." The model now has everything it needs.

ROLE AND PERSONA PROMPTING

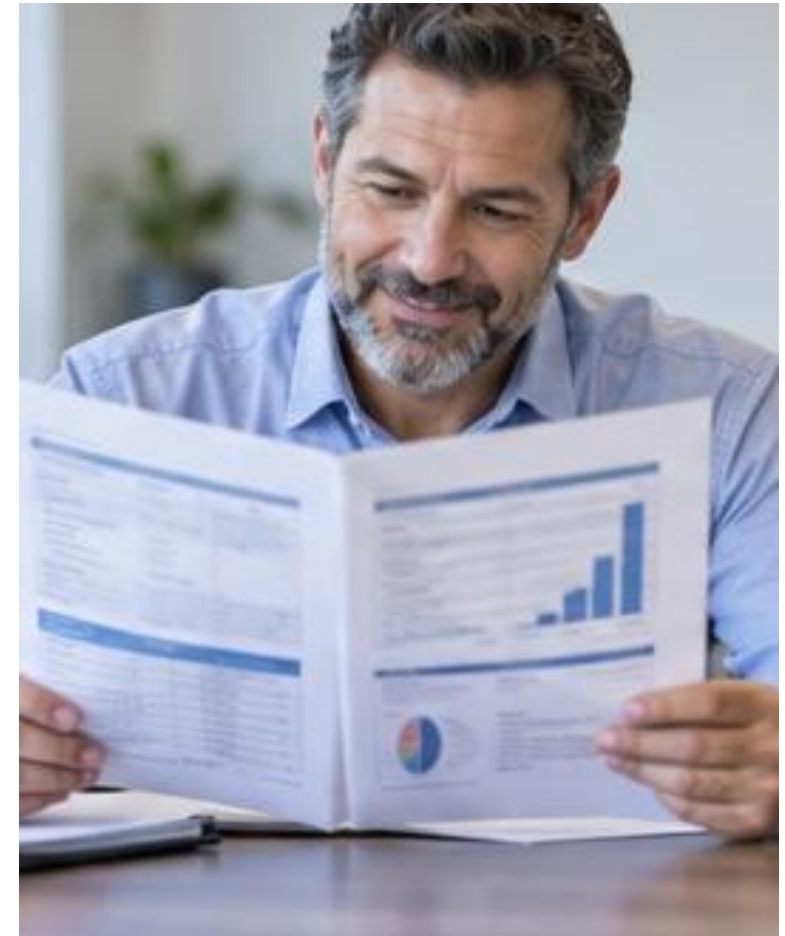
Telling ChatGPT to take on a role shapes the tone, level of detail, and assumptions it brings to the task. "You are an experienced financial analyst" produces different output than the same question with no role.

The role also sets the audience's expectations for what the response should contain. A legal reviewer, a copywriter, and a software architect will approach the same document very differently.

OUTPUT FORMAT AS A CONSTRAINT

Explicit format instructions prevent unwanted creative choices. If you need a markdown table, a JSON object, or a numbered list, say so.

This matters because ChatGPT will make a formatting choice if you do not. Sometimes that choice is helpful. Often it is not what you needed, and reformatting output wastes time.



FORMAT TYPES YOU CAN REQUEST

- Markdown: tables, headers, bullet lists, bold, code blocks
- Numbered lists with specific fields per item
- JSON or structured data with defined keys
- Prose paragraphs with explicit length limits
- Slide outlines with heading and sub-bullet structure
- Email, memo, or report templates with named sections

Markdown: Write Simple Text, Get Rich Formatting

Markdown lets you format text using easy symbols. It's widely used in documentation, chat, and the web.

1. Markdown Syntax (What You Type)

Add simple symbols to your text.

```
1 # Main Heading
2 ## Subheading
3
4 This is a paragraph with bold text and
5 italic text.
6
7 - Item 1
8 - Item 2
9 - Item 3
10
11 > This is a blockquote.
12
13 `Code` in a sentence.
14
15 ```
16 Code block (multiple lines)
17 ```
```



2. Rendered Output (What It Looks Like)

The same text, nicely formatted.

Main Heading

Subheading

This is a paragraph with **bold text** and *italic text*.

- Item 1
- Item 2
- Item 3

This is a blockquote.

`Code` in a sentence.

```
Code block
(multiple lines)
```

3. Teaching the Model to Use Markdown

You can tell the model to respond using Markdown. For example:



Your Prompt

Explain what Markdown is and give a short example. Use Markdown formatting.



Model Response (in Markdown)

```
# Markdown
Markdown is a lightweight way to format text.
Example:
- Use bold for emphasis
- Use italics for subtle emphasis
- Use code for code
> Keep it simple!
```



Quick Tips

- ✔ It's just text — easy to write and read.
- ✔ Great for notes, docs, and chatting with AI.
- ✔ Widely supported (GitHub, Notion, VS Code, etc.).
- ✔ Start with headings, lists, and code blocks.

ZERO-SHOT PROMPTING

Zero-shot means you describe what you want and let ChatGPT decide how to do it. There are no examples. You are relying on the model's training to infer the format and approach.

Zero-shot works well for common tasks where the output format is standard and variation is acceptable. Asking for "a professional email declining a meeting" is a zero-shot task most models handle well.

FEW-SHOT PROMPTING

Few-shot means you provide one or more examples of exactly the output you want before asking for more. The model learns the format, style, and classification logic from your examples.

Few-shot is the right choice when zero-shot results are inconsistent, when you are using a classification scheme the model would not know by default, or when the output format must match a very specific template.



ZERO-SHOT VS. FEW-SHOT: WHEN TO USE EACH

- Output format is standard and variation is acceptable: zero-shot
- You need exact format matching across many items: few-shot
- Classification uses your organization's own categories: few-shot
- Zero-shot has already been inconsistent or wrong: switch to few-shot
- Task is simple and self-explanatory: zero-shot first, upgrade if needed

STEERING VS QUEUING

Steering means guiding the model while it works: correcting direction, adding constraints, or refining the goal as new information appears. Steering can interrupt the agent while it is working, giving it new information before it finishes its response.

Queuing means giving the model a task and letting it run independently until it finishes, asks for help, or reports back.

Use steering when the work is ambiguous or evolving. Use queuing when the task is clear, repeatable, and can be completed without much back-and-forth.

BREAKING COMPLEX TASKS INTO STEPS

Complex tasks fail when given all at once because the model tries to optimize the entire response simultaneously. Breaking the task into sequential steps keeps each step focused and gives you a review point before moving forward.

This mirrors how you would delegate work to a human. You would not hand someone a one-page brief and expect a finished report. You would check in at the outline stage, the draft stage, and the revision stage.

WHY STEP-BY-STEP WORKS

When you ask ChatGPT to reason step by step, it produces fewer errors on tasks that require multiple logical dependencies. Each step can be validated before the next one is built on top of it.

This is especially important for analysis tasks where an early mistake compounds into a wrong conclusion. Catching the error at step two is far cheaper than catching it in the finished deliverable.



ITERATIVE REFINEMENT

The first response is rarely the final one. Refining within the same conversation is faster than starting over because the model retains context from all previous turns.

You do not need to re-explain the task, the audience, or the context on each follow-up. A targeted follow-up like "tighten the opening paragraph to two sentences" builds on what was already generated.

WHAT GOOD FOLLOW-UPS LOOK LIKE

Weak follow-ups restate the goal: "Make it better." Strong follow-ups are specific.

- "Replace the third paragraph with a numbered list of three risks."
- "Shorten the executive summary to 75 words."
- "Add one concrete example to support the claim in the second bullet."
- "Change the tone from formal to conversational throughout."
- "Remove any references to budget and reframe around timeline."



AI SELF-CRITIQUE

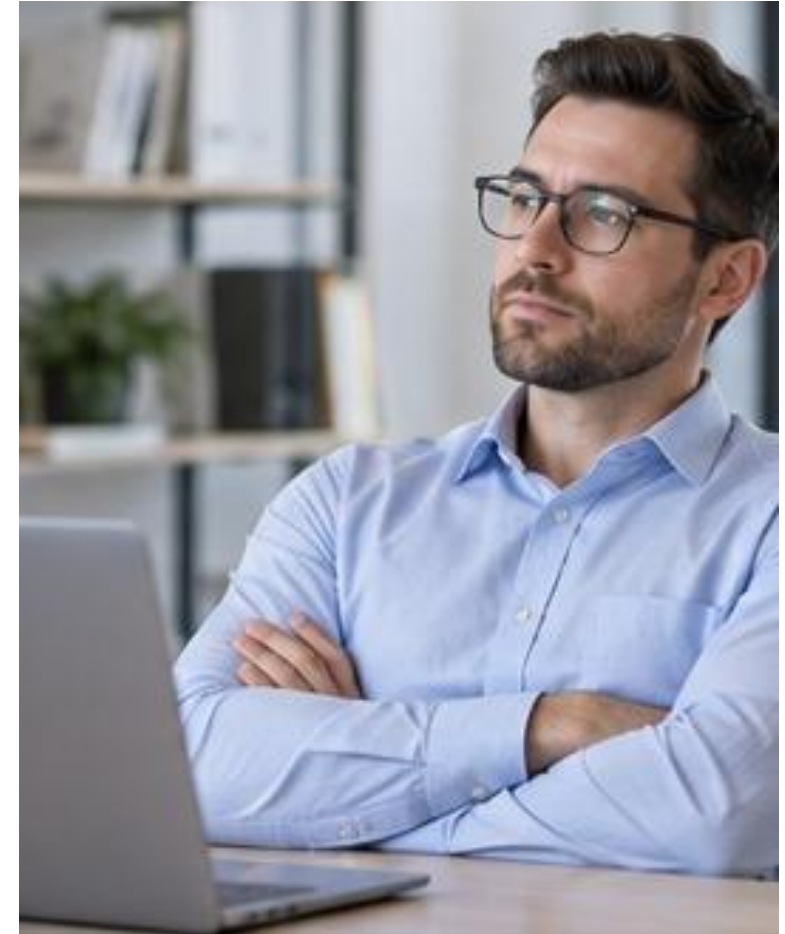
You can ask ChatGPT to evaluate its own output before you spend time editing it. Ask it to list assumptions it made, places where the response is vague or generic, and one specific improvement it would make if it rewrote the response.

This surfaces problems early and cheaply. It does not replace human review. Developing judgment about which critique to act on is itself a skill.

WHEN MORE PROMPTING STOPS HELPING

Not every gap in output quality can be fixed with a better prompt. Sometimes the model does not have the information it needs, the task requires real-world knowledge you have not provided, or the problem is genuinely beyond what this model handles well.

Recognizing when to stop and verify manually, add a document, or use a different tool is as important as knowing how to write a strong prompt.



PROMPT ENGINEERING: THE KEY PRINCIPLES

- A focused prompt beats a vague one, regardless of length
- Format is always worth specifying explicitly
- Few-shot anchors behavior when zero-shot varies
- Iterate in the same chat rather than starting over
- Use AI self-critique before spending time editing manually
- Step-by-step requests reduce errors on complex tasks
- Know when the problem is the prompt versus the task itself

PLAN MODE

Plan Mode lets you work with the model before it starts changing files, running code, or taking action. Instead of jumping straight into execution, the model helps break the task into steps, identify risks, and clarify what “done” should look like.

Plans are useful because they make complex work easier to inspect before committing to it. You can revise the plan, add constraints, remove steps, or change direction while the cost of changing your mind is still low.

Once the plan looks right, you can ask the model to execute it. The model then works through the agreed steps, updates you as it goes, and adjusts if new information appears.

PLAN MODE



LAB 1



POP QUIZ: A colleague sends ChatGPT "summarize the project" and receives a generic, unhelpful response. What is the most likely cause?

- a. ChatGPT is not capable of summarizing project information
- b. The model defaulted to a reasoning mode instead of conversational
- c. The prompt did not include a goal, context, format, or audience
- d. ChatGPT requires a file upload before it can summarize anything



MCQ 1 — ANSWER

Correct answer: c

The prompt did not include a goal, context, format, or audience. ChatGPT is capable of summarizing project information; the generic result comes from an underspecified prompt, not a model or tooling limitation.

POP QUIZ: An analyst needs to classify 50 customer emails into four internal priority categories. Zero-shot prompting has produced inconsistent results. What is the best next step?

- a. Rephrase the task description using simpler language
- b. Switch to a reasoning model for classification tasks
- c. Add one clearly labeled example per category to the prompt
- d. Use deep research to gather more context before classifying



MCQ 2 — ANSWER

Correct answer: c

Add one clearly labeled example per category to the prompt. Few-shot examples anchor the model's classification logic to your organization's specific categories, which is exactly what inconsistent zero-shot results call for.

WORKING BEYOND TEXT

ChatGPT can work with documents, images, spreadsheets, and web content in addition to generating text from instructions alone.

Knowing which capability to use for a given task determines whether you get a useful, grounded result or a hallucinated one. The wrong mode often looks plausible while being wrong.



FILE TYPES CHATGPT CAN READ

- PDF, Word, and text documents
- CSV, Excel, and structured data files
- PowerPoint and slide decks
- Images in common formats (JPEG, PNG, GIF, WebP)
- Code files in most programming languages
- Plain text and markdown files

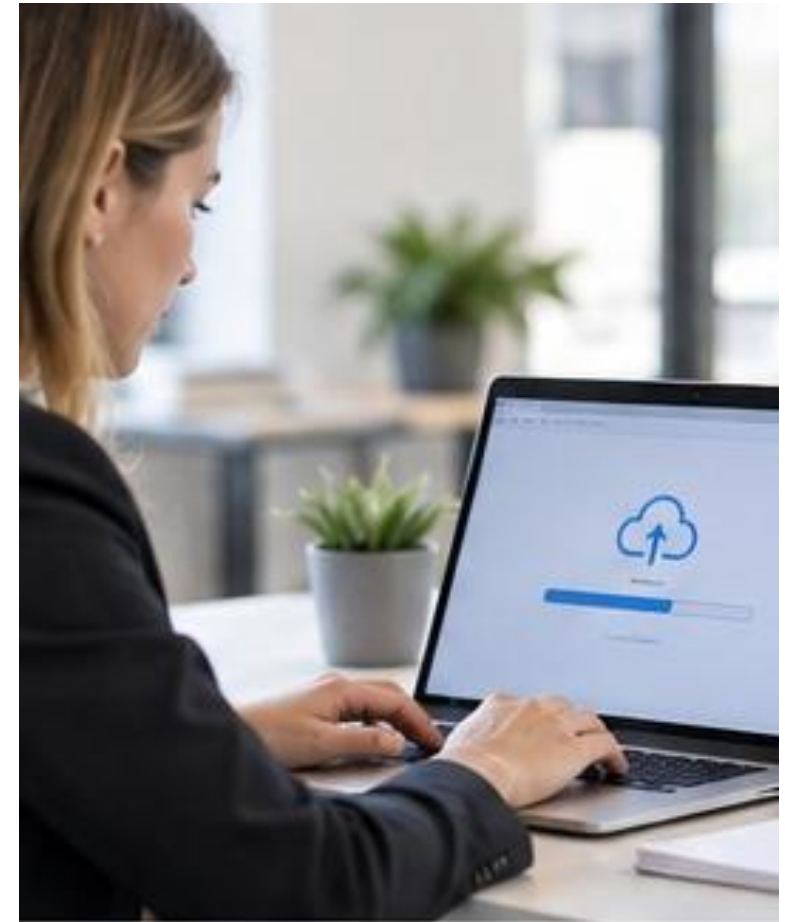
HOW DOCUMENT UPLOAD WORKS

When you upload a document, ChatGPT reads the content and uses it as the grounding source for your conversation. It answers questions, extracts information, and reasons over what is in the file.

Without an upload, ChatGPT has no knowledge of your internal documents. Uploading is the difference between AI working with your actual data and AI inventing answers about data it has never seen.

WHAT DOCUMENT UPLOAD IS GOOD FOR

- Answering specific questions about the document's content
- Extracting structured information (names, dates, metrics) into tables or lists
- Summarizing long documents at varying levels of detail
- Comparing two uploaded documents side by side
- Converting content from one format to another (prose to bullets, report to memo)
- Finding inconsistencies or gaps within a document



Choosing the right ChatGPT mode

Pick the mode based on where the answer has to come from. The wrong mode produces confident work from the wrong evidence.

Read files

Uploaded docs, spreadsheets, images

Use when the truth lives in your materials.

Risk: misses buried details if files are broad.

Search web

Current public information

Use when freshness or external sources matter.

Risk: weak sources look credible unless checked.

Calculate

Code, math, tables, transformations

Use when numbers or repeatable logic matter.

Risk: bad input data still gives precise output.

Generate

Model training and your prompt context

Use for drafting, reframing, examples, ideation.

Risk: plausible wording can mask uncertainty.

Decision rule

If the answer must quote a file, use file reading. If it must be current, use search. If it must compute, use code. If it must shape language or explore options, generate first and then verify.

Many real workflows chain modes: read the file, search for benchmarks, calculate the comparison, then generate the final deliverable.

IMAGE ANALYSIS

ChatGPT can interpret screenshots, charts, diagrams, photos, and handwritten notes. It can describe what it sees, extract visible text, summarize data, and explain processes shown in diagrams.

Image analysis works by examining the visual content directly. No file format conversion is required. You upload the image and describe what you want to know.

CHATGPT MEMORY

ChatGPT can remember useful information from previous conversations and use it to personalize future responses.

Account memories

Stored on OpenAI's servers

Available across devices when signed into the same account

View, delete, or disable them under **Settings** → **Personalization** → **Memory**

Local memories

Optional memory stored on the computer running the desktop app

Disabled by default

Can be enabled in the desktop app's settings

Key point: Memories provide additional context to the model. They do not retrain or permanently modify the underlying model.

IMAGE ANALYSIS: CAPABILITIES AND LIMITS

Strong use cases:

- Describing a diagram or process flow from a screenshot
- Extracting visible text from a photo or whiteboard image
- Identifying trends in a simple chart

Use with caution:

- Precise numeric values from complex charts (verify against the source)
- Small text or low-resolution images (transcription may have errors)
- Identifying specific people or private logos (results may be inaccurate)

DATA ANALYSIS AND THE CODE INTERPRETER

When you upload a CSV or spreadsheet, ChatGPT does not guess at the numbers. It uses its code interpreter to run actual Python code on your data.

Totals, averages, filtered results, and visualizations are computed from the file you provided. The output is as accurate as the data you uploaded.



WHAT THE CODE INTERPRETER CAN DO

- Sum, average, filter, and group records from structured data
- Identify trends and outliers across datasets
- Produce bar charts, line graphs, scatter plots, and pie charts
- Clean and reformat data (fix headers, fill blanks, remove duplicates)
- Run calculations across thousands of rows instantly
- Export results in different formats

GENERATED IMAGES

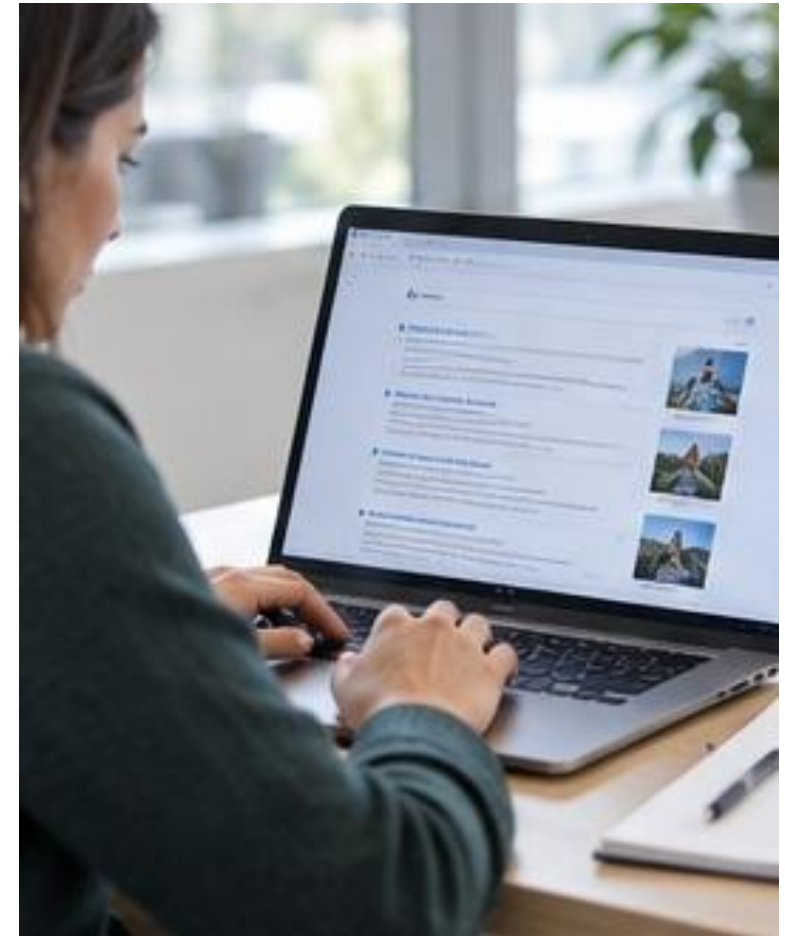
ChatGPT can generate original images from a text description. This is useful for drafting visual concepts, creating illustrations for documents, and exploring design directions without involving a designer.

Generated images are not photographs. They are AI-created visuals based on your description. They can be iterated on through follow-up prompts, just like text.

WEB SEARCH IN CHATGPT

When web search is enabled, ChatGPT can retrieve current information from the internet and include it in its response. Sources are cited inline.

Web search does not replace ChatGPT's underlying knowledge. It supplements it with information that is more recent than the training cutoff or more specific than general training data contains.



WHEN TO SEARCH VS. WHEN TO KNOW

- Current events, prices, or news from the past year: use web search
- Standard facts, general knowledge, well-established concepts: ChatGPT knows
- Company or product information that changes frequently: use web search
- Your organization's internal data: upload the document
- Regulatory guidance published after training cutoff: use web search

CHOOSING THE RIGHT MODE: A DECISION GUIDE

- Is the answer in a document you have? Upload it.
- Does the task involve math or data on many rows? Upload as CSV.
- Is the information from the past year or highly specific? Use web search.
- Is it a broad research task with many sources? Use deep research.
- Is it a writing, drafting, or generation task? Use conversational mode.
- Does it require sequential logic with many steps? Use a reasoning model.

MULTI-DOCUMENT AND MULTI-MODAL WORK

You can upload multiple files in a single conversation and ask ChatGPT to work across all of them. It can compare a contract against a policy, analyze a report alongside a spreadsheet, or combine a screenshot with a document.

This is particularly powerful in research workflows where you are synthesizing internal data with external information at the same time.

LAB 2



POP QUIZ: You upload a quarterly sales spreadsheet and ask ChatGPT to calculate total revenue by region. What is actually happening under the hood?

- a. ChatGPT estimates totals based on patterns from its training data
- b. ChatGPT searches the web for matching revenue figures to compare
- c. ChatGPT runs Python code on your file using its built-in code interpreter
- d. ChatGPT reads your screen and your open spreadsheet application in real time



MCQ 3 — ANSWER

Correct answer: c

ChatGPT runs Python code on your file using its built-in code interpreter. Totals and breakdowns are computed directly from the uploaded data, not estimated or looked up.

POP QUIZ: You ask ChatGPT to analyze a bar chart screenshot and extract the exact revenue figure for each region. How should you treat the output?

- a. Accept the numbers fully; ChatGPT reads pixel values with high precision
- b. Use the output as a starting point and verify figures against the source data
- c. Reject the output entirely; ChatGPT cannot interpret bar charts
- d. Only trust the output if you also upload the underlying spreadsheet



MCQ 4 — ANSWER

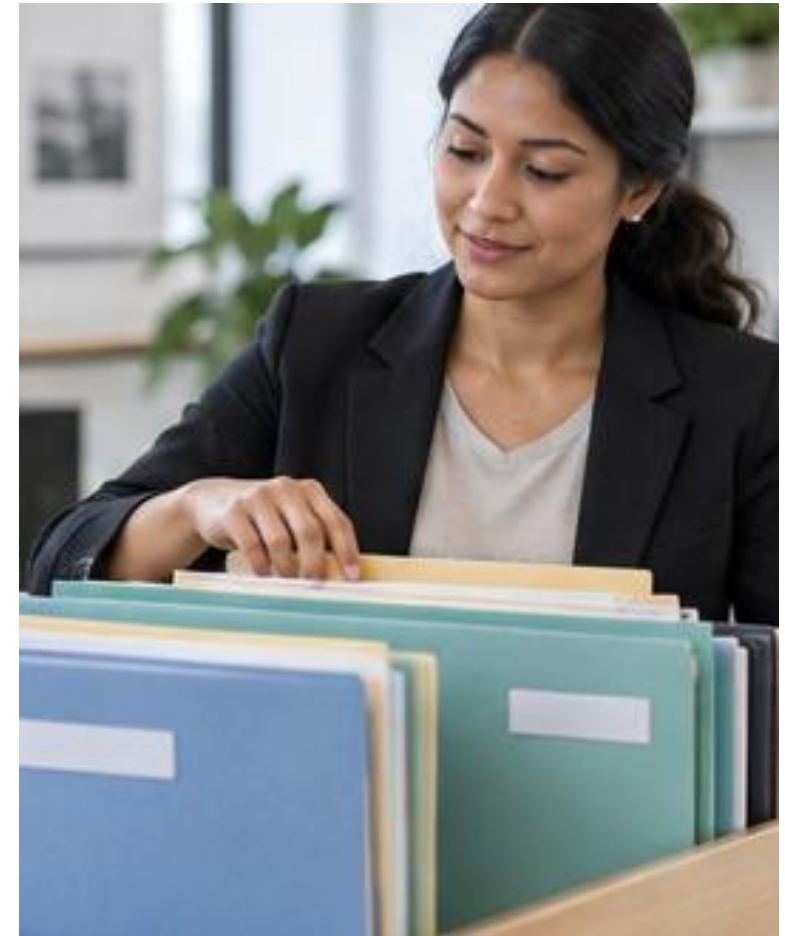
Correct answer: b

Use the output as a starting point and verify figures against the source data. Image analysis can misread precise numeric values from charts, so treat extracted figures as estimates to confirm.

CHATGPT PROJECTS

A Project is a persistent workspace in ChatGPT. The project instruction, uploaded files, and conversation history all stay associated with the project across sessions and across team members.

You write the project instruction once. Every conversation inside that project starts with the full context already loaded, with no manual setup required.



WHAT A PROJECT CONTAINS

- A project instruction that sets the context and output standards for every conversation
- Uploaded files that are accessible in all conversations inside the project
- Individual conversations (which can be organized by task or date)
- Notes and research outputs saved within the project workspace

WRITING EFFECTIVE PROJECT INSTRUCTIONS

A project instruction is not a prompt. It is a standing brief that shapes every conversation inside the project from the first message.

Write it to answer: who is the user, what is this project for, what should every output look like, what sources or data should always be referenced, and what terminology or framing matters to this team.

STRONG VS. WEAK PROJECT INSTRUCTIONS

A weak instruction says: "Help me with customer research."

A strong instruction says: "You are supporting a strategy analyst at Apex Retail Group, a mid-sized retail chain with 85 locations in the Midwest. The purpose of this project is to evaluate AI-powered customer service platforms. All outputs should be professional and evidence-based. Cite sources for any statistics. Flag any information older than 18 months. Use the uploaded vendor shortlist and metrics file as the baseline for all analysis."



FILE MANAGEMENT IN PROJECTS

Files uploaded to a Project are available in every conversation inside that project automatically. You do not re-upload them each time.

This means a team can upload a single source of truth once and every analyst working in the project has access to the same baseline data without any coordination overhead.

MULTIPLE CONVERSATIONS, SHARED CONTEXT

Opening a second conversation inside a Project gives you a clean slate while retaining all the project instructions and uploaded files.

This is useful when you want to explore a different angle without cluttering the first conversation, or when you want to produce a different deliverable format from the same research base.

TEAM USE OF PROJECTS

Multiple people can work inside the same ChatGPT Project with a shared instruction. Everyone starts from the same context and output standards.

This replaces the informal process of emailing prompts around or explaining the background every time someone new joins a task. The instruction is the single source of truth.

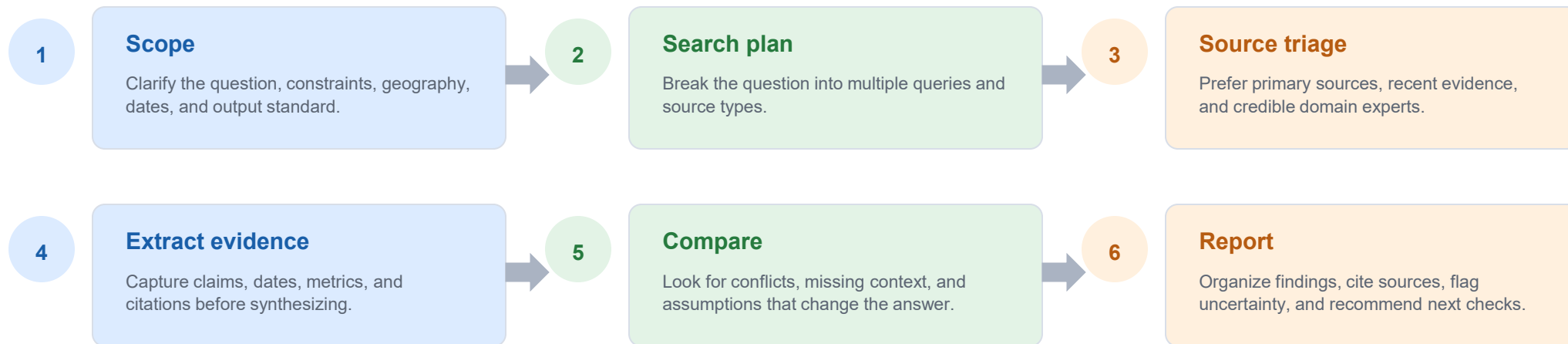


SEPARATING UNRELATED WORK

- Keep each project focused on one objective or domain
- Avoid mixing a client project with an internal operations project in the same workspace
- When a project is complete, archive it rather than repurposing it for something unrelated
- Personal tasks and team tasks should live in separate projects
- Context from one project should never bleed into another

How deep research works

Deep research is useful when the question needs source gathering, comparison, and synthesis rather than one quick lookup.



Instructor note

Deep research still needs a human checkpoint. Verify that the cited source says what the summary claims, and separate evidence from inference before using the output.

CRAFTING AN EFFECTIVE RESEARCH QUESTION

Open-ended questions with multiple legitimate answers are the right input for deep research. "What are the emerging risks in retail AI adoption and how are companies managing them?" is a good research question.

"What was the retail AI market size in 2023?" is a lookup. Use web search for that. Deep research is overkill for a single data point with a known answer.

OPEN-ENDED VS. LOOKUP QUESTIONS

- Lookup: "What is the SEC rule on AI disclosures?" — use web search
- Open-ended: "How is the SEC's approach to AI disclosure evolving and what are the compliance implications for advisory firms?" — use deep research
- Lookup: "Who are the top five retail AI vendors?" — use web search
- Open-ended: "What factors differentiate successful AI customer service deployments in mid-market retail from unsuccessful ones?" — use deep research

EVALUATING SOURCE QUALITY

Not all sources cited in deep research are equal. A government agency publication, a peer-reviewed study, and a vendor white paper carry very different levels of authority.

Train yourself to check three things before using a citation: who published it, when it was published, and whether the source actually says what the report claims it says.



CITATION RED FLAGS

- Source is a vendor white paper on their own product (potential bias)
- Publication date is more than two years old on a fast-moving topic
- No author or institutional affiliation listed
- The URL cited no longer loads or has been redirected
- The source is cited but the specific claim does not appear in the source
- Multiple claims trace back to a single secondary source that itself lacks primary citations

WHEN SOURCES DISAGREE

Deep research will sometimes surface contradictory findings. One study says AI customer service improves CSAT by 20%. Another says it has no measurable effect.

Do not let ChatGPT paper over this with a synthesized average. Flag the disagreement explicitly in your deliverable and note that the evidence is mixed. Decision-makers need to know where the data is uncertain.

WHAT TO VERIFY BEFORE USING RESEARCH

- Any specific statistic that appears without a verifiable primary source
- Claims from sources published more than 18 months ago on evolving topics
- Any assertion described as "recent" or "current" without a specific date
- Quoted language attributed to a specific person or document
- Regulatory guidance that may have been updated since the source was written



YOUR PERSONAL DECISION FRAMEWORK

After comparing web search and deep research on the same question, you will develop a sense for when the investment of time is worth it.

The key variable is not the topic. It is the stakes. A quick background check before a call does not need deep research. A recommendation memo that will go to senior leadership does.

COMBINING INTERNAL AND EXTERNAL INFORMATION

The most powerful research setup combines your internal documents with external research in the same Project. ChatGPT can reference your proprietary data while also pulling in current public information.

This is what makes research output relevant to your specific situation rather than generic. Industry trends matter more when they are anchored to your organization's actual numbers.

FROM RESEARCH TO DELIVERABLE

Ask ChatGPT to convert research findings into a structured business deliverable in a single step. A brief, a memo, or a risk summary can be drafted directly from the research output.

The draft will require human review. The research-to-draft step eliminates blank-page time and organizes the evidence. It does not eliminate the need for judgment or accountability.

LAB 3



POP QUIZ: Your team needs a structured analysis of data privacy regulations across five countries over the past three years. Which tool is the best fit?

- a. Web search, because it is faster and already cites sources inline
- b. Document upload, because your team should supply the regulations directly
- c. Deep research, because the task requires breadth, depth, and source synthesis
- d. Standard chat with no web access, because ChatGPT knows this topic from training



MCQ 5 — ANSWER

Correct answer: c

Deep research, because the task requires breadth, depth, and source synthesis. Comparing regulations across five countries and three years needs many searches and cross-referencing that only deep research is built for.

POP QUIZ: ChatGPT's deep research output cites a statistic about retail AI adoption from a trade publication. Before including it in a client report, what should you do?

- a. Include it without checking; deep research sources are automatically verified
- b. Ask ChatGPT to confirm the statistic by running another search on the same topic
- c. Check that the cited source actually contains the specific statistic being claimed
- d. Paraphrase the statistic so you no longer need to include a citation



MCQ 6 — ANSWER

Correct answer: c

Check that the cited source actually contains the specific statistic being claimed. Citations can be mismatched to a claim, so the only reliable check is reading the source directly.

BUILDING AN AI RESEARCH WORKSPACE

A research Project combines a clear instruction that defines the purpose, uploaded internal documents as a knowledge base, and web search for external information.

The result is a workspace where ChatGPT understands your context before you ask a single question. You spend your time on the research, not on re-explaining who you are.



THE PROJECT INSTRUCTION FOR RESEARCH WORK

- State who you are and the organization's context
- Name the specific evaluation or research objective
- Specify output standards (professional, evidence-based, sources cited)
- Tell ChatGPT which uploaded files to treat as the internal baseline
- Define what to flag (outdated information, unverified claims, missing sources)

GROUNDING CHATGPT IN YOUR DATA FIRST

Before asking ChatGPT to bring in external research, ask it to summarize what it knows from your uploaded files. This confirms the project files are accessible and that the baseline is correct before you add outside information.

Starting with internal context and then layering in external research produces more relevant output than doing both at once.

COMBINING INTERNAL AND EXTERNAL RESEARCH

Ask ChatGPT to bring in external benchmarks or competitive data only after it has established the internal baseline.

For example: first confirm what Apex Retail's current customer service metrics look like from the uploaded file, then ask for industry benchmarks to compare against. The comparison is only meaningful when both sides of it are accurate.



STRUCTURING THE RESEARCH CONVERSATION

- Start each conversation with a context-confirming question
- Work through research topics one at a time, not all at once
- After each major finding, ask: is there anything that contradicts this?
- Review output before moving to the next research step
- Keep different research angles in separate conversations within the same project

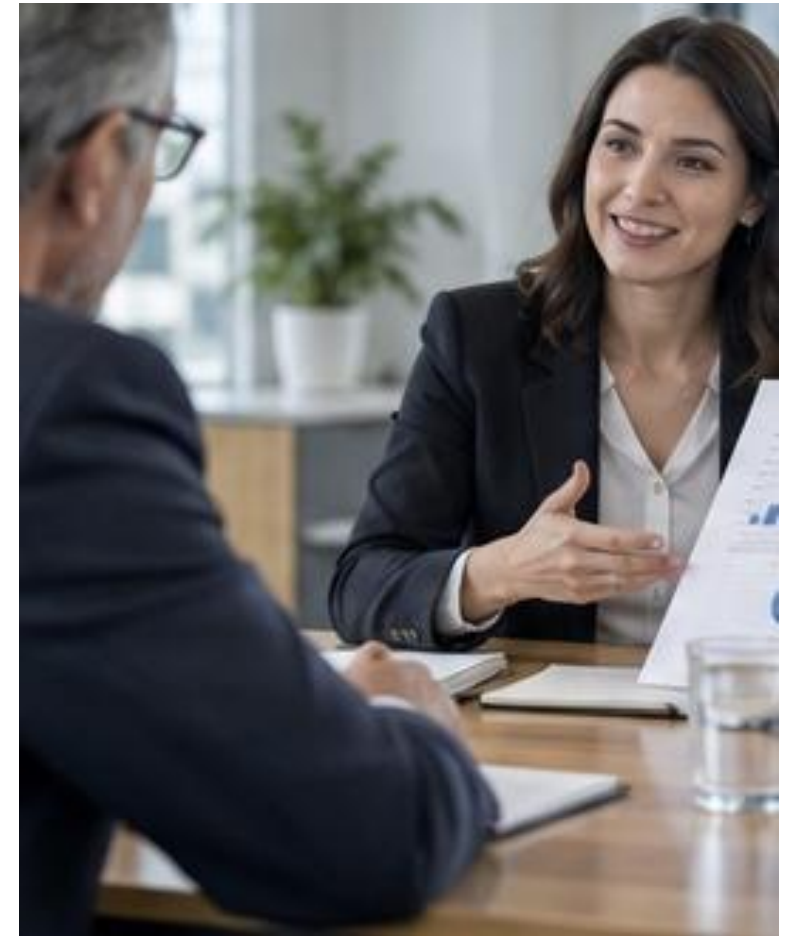
THE COMPOUNDING VALUE OF PROJECTS

Every conversation inside a well-configured Project starts with full context. You do not re-explain who you are, what the output style is, or which files are relevant.

As the project accumulates more conversations, each new one becomes more focused and productive with less setup. The upfront investment in a strong project instruction pays off with every conversation that follows.

WHAT A STRONG DELIVERABLE LOOKS LIKE

- A clear recommendation or conclusion stated at the top, not buried at the end
- Evidence cited by source, not described as "research suggests"
- Claims that go beyond the evidence flagged explicitly as assumptions
- A clear statement of what additional information would change the recommendation
- Formatted for the audience, not for ChatGPT's default style



THE CRITIQUE STEP

After ChatGPT produces a draft deliverable, ask it to critique that draft. Ask it to identify any claims not supported by the research, any sections where more information would strengthen the argument, and any assumptions it made that you did not ask it to make.

Review the critique before deciding which changes to make. Not every critique will be valid. Some will be stylistic preferences. Others will surface real gaps you should address.

HUMAN CHECKPOINTS ARE NOT OPTIONAL

- Someone with domain knowledge must verify that the sources say what the output claims
- The conclusions must follow logically from the evidence, not just sound plausible
- Any claim that will be attributed to your organization requires human sign-off
- The person accountable for the deliverable must read the full output before it is sent
- AI can draft and organize. Humans remain responsible for accuracy and judgment.

DAY 1 KEY TAKEAWAYS

- LLMs generate text statistically. They do not look facts up. Always verify specific claims.
- Structured prompts with goal, context, format, and audience outperform vague ones consistently.
- Use the right mode: upload for internal data, web search for current info, deep research for complex synthesis.
- Projects eliminate repeated setup and give teams a shared context baseline.
- Human checkpoints are not a nice-to-have. They are the control that makes AI output trustworthy.

LAB 4



POP QUIZ: What is the main advantage of uploading files to a ChatGPT Project compared to attaching them in a one-off conversation?

- a. Project files are indexed by search engines for faster retrieval
- b. Files uploaded to Projects are processed faster than files in regular chats
- c. Project files apply automatically to every conversation started inside that project
- d. ChatGPT stores Project files permanently and shares them across all accounts



MCQ 7 — ANSWER

Correct answer: c

Project files apply automatically to every conversation started inside that project. That is the specific advantage over one-off attachments, which only apply to a single conversation.

POP QUIZ: A new team member opens an existing ChatGPT Project for the first time. What does the project instruction provide them?

- a. A complete transcript of all previous conversations in the project
- b. A shared starting context for new conversations without needing to re-explain anything
- c. Automatic web search enabled for every conversation they start in the project
- d. A locked set of prompt templates that cannot be modified by individual users



MCQ 8 — ANSWER

Correct answer: b

A shared starting context for new conversations without needing to re-explain anything. The instruction is the standing brief, not a transcript, a search toggle, or a locked template set.

CONGRATS!



Today covered the foundations of working with ChatGPT.

How LLMs work: tokens, context windows, nondeterminism, and hallucinations.

Prompt engineering: structure, roles, output formats, and iterative refinement.

Beyond text: file uploads, image and data analysis, and web search.

Projects and deep research: shared context and source evaluation.